Day 8: Shape patterns, and the 10th step without drawing

Name _____ Date _____

I can follow a rule to make a shape pattern. I can find a later step without drawing.

Warm-up.

Find all the factor pairs of 82.	**Is 97 a multiple of 3? Show how you know.**	**Is 78 a multiple of 6? Show how you know.**

Fix it 1. Kai made start at 3, multiply by 2: 3, 6, 12, 24. He looked at 3 and 6. He says odd and even take turns.

What went wrong: _____

Something true of the numbers: _____

Example 1. Step 1 is a column of 2 squares. Each step adds one more column of 2. How many squares are in the 10th step? What is true that the rule did not say? Write on your recording page, shape pattern 1.

The 10th step is the start and _____ jumps: _____ + _____ = _____

I notice _____

Tapes. Write on your recording page, the shape tapes.

Tape 1. Start at 3 tiles. Add 2 tiles each step. The tape shows 3, 5, 7. Write the next five. How many tiles are in the 10th step?

The 10th step: _____ tiles

Tape 2. Start at 2 tiles. Add 5 tiles each step. The tape shows 2, 7, 12. Write the next five. How many tiles are in the 10th step?

The 10th step: _____ tiles

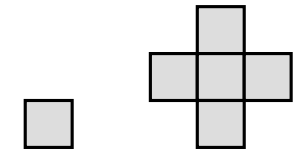
Fix it 2. Jo made stick squares in a row: 4, 7, 10, 13, 16 sticks. For the 10th step, Jo drew all ten squares and counted 31 sticks.

Jo is right. What is slow about Jo's way? _____

A faster way: _____

On your own.

Plus sign. A plus sign grows. Step 1 is 1 square. Each step adds 4 squares, one at the end of each arm. How many squares are in the 10th step? Write one thing the rule did not say. Why will it keep going? Write on your recording page, shape pattern 3.



Step 1 Step 2

I notice _____

It will keep going because _____

_____

Shapes that repeat. A pattern repeats circle, square, triangle. Draw the first 12 shapes. Which shapes are squares?

Squares: _____ _____ _____ _____

Written task. The triangles in circle, square, triangle are the 3rd, 6th, 9th and 12th shapes. Sam says the 30th shape is a triangle. Is Sam right? Write why.

_____

_____

_____